

Correlation

1. From the following table, calculate the coefficient of correlation by Karl Pearson's method

X	6	2	10	4	8
Y	9	11	5	8	7

- 2.. Calculate the coefficient of correlation from the following

X	9	8	7	6	5	4	3	2	1
Y	15	16	14	13	11	12	10	8	9

- 3.. Calculate Pearson's coefficient of correlation for the following series:

Price (Rs)	22	24	26	28	30	32	34	36	38	40
Demand (Tons)	60	58	58	50	48	48	48	42	36	32

4. Find out the coefficient of correlation for the following data.

Husband's Age	23	27	28	29	30	31	33	35	36	39
Wife's Age	18	22	23	24	25	26	28	29	30	32

5. Find Karl Pearson's co-efficient of correlation between the values of X and Y given below and calculate the probable error.

X	78	89	96	69	59	79	68	61
Y	125	137	156	112	107	136	123	108

Assume 69 and 112 as the mean values for X and Y , respectively.

6. Compute Karl Pearson's coefficient of correlation between exports and imports in crores of Rs. from the following data:

Year	1987	1988	1989	1990	1981	1992	1993
Exports	115	118	122	120	126	127	125
Imports	140	138	142	146	145	148	146

7. The following table gives the distribution of the total population and those who are wholly or partially blind among them.

Find out if there is any relation between age and blindness.

Age	No. of Persons ('000)	Blind	Age	No. of Persons ('000)	Blind
0–10	100	55	40–50	24	36
10–20	60	40	50–60	11	22
20–30	40	40	60–70	6	18
30–40	36	40	70–80	3	15

8. Find the correlation coefficient between age and playing habits of the following students:

Age	15	16	17	18	19	20
No. of Students	250	200	150	120	100	80
Regular Players	200	150	90	48	30	12

What conclusion can be drawn from the result?

9. Find the coefficient of correlation from the following data :

x	6	2	10	4	8
y	9	–	5	8	7

Arithmetic means of x and y are 6 and 8, respectively.

10. A student calculates the value of r as + 0.72 for a question comprising 5 pairs of observations and concludes that there is high degree of correlation. Justify his conclusion by calculating the probable error.
11. A survey regarding the incomes and savings of 100 school teachers in a certain city provided the following data:

Income (Rs)	Saving (Rs)				
	50	100	150	200	Total
400	8	4	–	–	12
600	–	12	24	6	42
800	–	9	7	2	18
1000	–	–	10	5	15

1200	–	–	9	4	13
Total	8	25	50	17	100

Compute the correlation coefficient between income and savings.

12 Calculate r from the following data by Karl Pearson's method.

Test whether this value

of r is significant or not:

X Y	0	1	2	3
15	10	–	13	–
20	–	12	–	–
25	11	–	15	–
30	–	14	–	5

13. Family income and its percentage spent on food in case of 100 families gave the following bivariate frequency distribution. Find if there is any relationship.

Food expenditure %	Family income (Rs.)				
	200–300	300–400	400–500	500–600	600–700
10	—	—	—	3	7
15	—	4	9	4	3
20	7	6	12	5	—
25	3	10	19	8	—

- 14 From the data given below, find the whether there is any relationship between the age of the drivers and the number of accidents made by them.

Driver's age	No. of accidents				
	0-2	2-4	4-6	6-8	8-10
25-30	—	—	3	4	12
30-35	3	—	5	9	7
35-40	5	9	10	6	3
40-45	7	4	3	—	1
45-50	8	1	—	—	—

(T.U. 2067)

15. The following information is related to the ages of husband (x) and wife (y) for married couples living together in a sample survey. Calculate the coefficient of correlation between the husband's and his wife's ages. Test the significance of the calculated r.

$$N = \sum f = 72, \sum fx = 3560, \sum fx^2 = 196800, \sum fy = 3260,$$

$$\sum fy^2 = 168400 \text{ and } \sum fxy = 172000 \quad (T.U. 2061)$$

16. Calculate the coefficient of correlation between X and Y series from the following data:

$$n = 15, \bar{X} = 25, \bar{Y} = 18, \sigma_x = 3.01, \sigma_y = 3.03,$$

$$\sum (X - \bar{X})^2 = 136, \sum (Y - \bar{Y})^2 = 138$$

$$\sum (X - \bar{X})(Y - \bar{Y}) = 122$$

Regression

- 1.(a)The following table gives the age of cars of a certain make and annual maintenance costs.

Obtain the regression equation for costs related to age:

Age of cars (years)	2	4	6	8
Maintenance cost (Rs.'00)	10	20	25	30

Estimate the maintenance cost for a ten-year old car.

- (b) The advertisement expenses and the sales of a new product are recorded as below :

Adv. exp. (Rs '000)	1	5	6	8	10
Sales (Rs '000)	50	60	80	100	100

Estimate the sales when the advertising expenses is Rs 15,000

(T.U. 2063)

2. The following table gives the aptitude test scores and productivity indices of 10 workers selected at random:

Aptitude Scores X	60	62	65	70	72	48	53	73	65	82
Productivity index Y	68	60	62	80	85	40	52	62	60	81

Calculate the two regression equations and estimate the productivity index of a worker whose test score is 92.

3. The following data gives the experience of machine operators in years and their performance as given by the number of good parts turned out per 100 pieces.

Operator	1	2	3	4	5	6	7	8
Experience (X)	16	12	18	4	3	10	5	12
Performance (Y)	87	88	89	68	78	80	75	83

Calculate the regression line of performance ratings on experience and estimate the probable performance if an operator has 8 years experience.

4. Find the regression equations from the following data:

Age of husband	18	19	20	21	22	23	24	25	26	27
Age of wife	17	17	18	18	19	19	19	20	21	22

5. From the following data obtain the two regression equations and calculate the correlation coefficient:

X	1	2	3	4	5	6	7	8	9
Y	9	8	10	12	11	13	14	16	15

6. Find the two regression equations and the coefficient of correlation from the data given below:

Marks in statistics	40	38	35	42	30
Marks in maths	30	35	40	36	29

Also find the most likely marks in statistics when the mark in Mathematics is 34.

7. Calculate correlation and regression coefficients from the data:

X	2	4	6	8	10	12	14
Y	4	2	5	10	4	11	12

Find the estimate of Y when X = 13.

8. You are given below the information about advertising and sales:

	Advertising Expenditure (Rs. lakhs)	Sales (Rs Lakhs)
Mean	10	90
S.D.	3	12

The coefficient of correlation between advertising and sales = 0.8

- (a) Determine the two regression lines,

- (a) Consider the two regression lines $3x + 2y = 26$ and $6x + y = 31$

- Find the means and correlation coefficient between x and y.
- Show that the regression estimate of y for x = 0 is 13 where as regression estimate of x corresponding to y = 13 is 3. Explain the cause of difference. ($r = -0.5$)
- Regression coefficient of Y on X and X on Y are given as : -2.002 and -0.461 . Find the value of correlation coefficient between X and Y. If mean of X and Y are 87.2 and 127.2 respectively, estimate X when Y = 133. (T.U. 2065)

(b) In a regression study, the two regression lines are $2X - 3Y + 6 = 0$ and $4Y - 5X - 8 = 0$. Calculate the means of X and Y. If the standard deviation of X is 3, determine the standard deviation of Y.

9. To study the relationship between expenditure on accommodation X and expenditure on food and entertainment Y, an enquiry into 50 families gave the following results :

$$\sum X = 8,500, \sum Y = 9,600$$

$$\sigma_X = 60, \sigma_Y = 20 \text{ and } r = 0.6$$

Estimate the expenditure on food and entertainment when accommodation costs Rs. 200.